CREATIVE-TECHNICAL PORTFOLIO / PROJECT 03

OUTDOOR POLICE BODY-WORN CAMERA RECONSTRUCTION

EGOCENTRIC VIDEO / DYNAMIC MOTION

Three short outdoor reconstructions examining movement-linked camera pose, pursuit-speed viewpoint changes, wide-angle imaging and spatial-temporal video quality.



03 INDEPENDENT CLIPS

18.1 s COMBINED DURATION

THE MOVING CAMERA IS THE CENTRAL PRODUCTION CHALLENGE

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Multimodal Visual Direction | Generative Video Production | Output Evaluation

AI-generated bodycam-style reconstructions; not authentic police recordings.

EXECUTIVE OVERVIEW

A moving viewpoint as the primary subject

This independent creative-technical study evaluates whether generated video can reproduce recognizable outdoor body-worn camera behavior across three distinct residential scenarios.

THE WORK

I developed three approximately six-second bodycam-style clips. Two explore outdoor residential movement and scene coverage; the third presents a faster pursuit-style sequence in which the viewpoint follows a running figure along a sidewalk and toward a front yard. All three are independent visual outputs, not continuous footage from one incident.

Production contribution

Visual direction for first-person camera origin, relative camera motion, field-of-view treatment, outdoor scene geometry, action pacing and selection of representative generated results.

Review contribution

Qualitative evaluation of motion plausibility, subject visibility, built-environment stability, viewpoint transitions, recording-overlay presentation and continuity within each clip.



Source frame / Clip 03: the viewpoint follows a running figure from the pavement into a front-yard approach.

PROJECT SCOPE

The purpose is to demonstrate controlled simulation of a camera attached to a moving wearer, not to claim the recreation of a recorded police operation or to establish a real-world incident.

SOURCE-FOOTAGE REVIEW

Three clips, three movement conditions

A comparative study is valuable only when the visual evidence is described accurately. The first two clips show residential movement and reorientation; the third contains the clearest running sequence.

01 / OPERATIONAL STREET VIEW



An overcast residential street with multiple uniformed figures. A foreground figure enters from the right as the camera heading shifts; bodycam-style framing and overlays remain visible.

02 / RESIDENTIAL WALK-THROUGH



A forward-looking residential sidewalk view. The camera pans or yaws with simulated wearer movement while houses, parked vehicles, roadway and tree remain identifiable.

03 / RUNNING PURSUIT VIEW



A runner moves ahead of the camera past parked cars, then toward a lawn and residential frontage. The faster movement produces greater viewpoint rotation and image displacement.

TECHNICAL PROBLEM

From stable framing to wearer-linked motion

The defining constraint is not that the camera shakes. It is that apparent camera translation and rotation remain causally connected to the movement of a person wearing the device.

Camera pose

A body-worn camera has a moving pose: its viewpoint changes in position and orientation as the wearer steps, pivots, leans or runs. Yaw, pitch and roll variation should be compatible with that movement.

Scene response

The sidewalk, parked vehicles, houses and trees shift in image coordinates as viewpoint and viewing direction change. These relationships distinguish ego-motion from objects moving independently.



Clip 03 / 00:02.4: the runner remains in view as the camera closes distance along the sidewalk.

THE DESIGN DECISION

For a running shot, cadence-related vertical oscillation, lateral sway and brief horizon rotation can support realism. Smooth, stabilized tracking would describe a different camera system. Uncontrolled jitter, however, is not a substitute for physically coherent motion.

EVIDENCE BOUNDARY

The evaluation therefore links the apparent shake to motion direction, path changes and changes in visual landmarks. It does not infer measured body acceleration, camera mount dimensions or actual stabilizer settings from the generated pixels.

DYNAMIC CAMERA BEHAVIOR

Running-induced vibration and viewpoint change

Clip 03 is the project's strongest example of higher-motion egocentric capture: the camera follows a figure at speed as the path redirects from pavement toward a house.



00:00.8 / approach



00:02.8 / trajectory change



00:04.8 / lawn entry

Translational motion

Forward travel changes apparent scale and proximity of parked vehicles, the moving person and nearby houses. The scene does not read as a static camera with an independently animated subject.

Rotational motion

The visual axis reorients as the moving figure changes direction. The apparent horizon and near-edge geometry change with that reorientation rather than remaining level throughout.

CAPTURE-STYLE OBSERVATION

Visible motion blur and edge displacement increase in parts of the running clip. This is consistent with the intended high-movement aesthetic, but the source video alone cannot establish the precise physiological motion or sensor parameters that produced the effect.

CORE FINDING

The most convincing evidence is the coupling among forward movement, changing frame composition and the runner's path - not the presence of shake considered in isolation.

PURSUIT TRAJECTORY

Spatial continuity through a route change

As the camera follows the subject, maintaining legible movement requires that the street, footpath, parked cars, lawn and house retain coherent relative positions despite the rapidly changing view.



Source frame / Clip 03: the path is still sidewalk-aligned, with street parking on the left and houses beyond.



00:03.6 / bend toward lawn



00:05.6 / residential frontage

OBSERVED SEQUENCE PROGRESSION

The clip moves from a forward sidewalk view to a leftward/yardward trajectory near the house frontage. The moving subject remains readable during this transition, while parked cars and residential structures provide spatial context.

STORYBOARD AND EDITORIAL VALUE

A production review checks whether the subject appears to follow a traversable route and whether background geometry responds continuously to the camera turn. No arrest, confrontation or factual case outcome is implied by the imagery.

OUTDOOR OPERATIONAL FRAMING

Reading an overcast street at human height

Clip 01 emphasizes operational spacing and a moving first-person viewpoint among uniformed figures, without relying on the sprint-level dynamics of the third clip.



Clip 01 / 00:01.6: wide street coverage and visible figures near the roadside establish relative position and approach context.



00:03.6 / wearer approaches



00:05.2 / foreground figure

Human-scale framing

The sidewalk, grass verge and uniformed figure entering close to the lens provide a clear near-field reference, distinguishing on-foot perspective from an elevated fixed-install camera.

Motion and visibility

Reorientation retains the general street layout, while the foreground figure changes in scale and position. The clip illustrates person-to-person spacing and pass-by framing rather than an uninterrupted running chase.

RESIDENTIAL MOVEMENT STUDY

Viewpoint exploration in a built environment

Clip 02 offers a lower-intensity comparison: the viewing direction changes as the scene moves along a neighborhood sidewalk under overcast daylight.



Clip 02 / 00:00.8: initial alignment down the sidewalk, with road vehicles and house fronts as depth cues.



00:03.2 / viewpoint reorientation



00:05.6 / later street coverage

OBSERVABLE STRENGTH

Residential façades, tree position, sidewalk segments and a parked silver car support spatial orientation as the camera pans and advances. This makes the clip useful for evaluating scene persistence and the visual effect of steering a body-mounted view.

TECHNICAL DISTINCTION

The moderate apparent motion differentiates this clip from the pursuit shot. Its contribution is broad environmental coverage and a plausible first-person view, not evidence of high-amplitude running.

OPTICAL AND IMAGE CHARACTERISTICS

Wide-angle coverage and image-edge behavior

The clips depict broad field coverage with substantial peripheral darkening and a circular image boundary. These are observable rendering characteristics, not measurements of a physical lens.



Clip 03: the image conveys broad horizontal coverage, edge curvature and peripheral darkening while a runner remains visible in the scene.

Field coverage

A wide angle makes sidewalks, parked cars, houses and a moving figure visible within one view. It also creates noticeable near/far scale differences, especially as the camera passes vehicles.

Peripheral image treatment

A curved image boundary, vignetting and some edge stretching produce an action-camera/bodycam aesthetic. They should be evaluated for consistency, not treated as measured distortion coefficients.

TECHNICAL CLAIM STANDARD

The footage establishes appearance, not optical hardware specifications. Lens field-of-view angle, focal length, sensor type and stabilization method are not independently recoverable from the rendered source clips.

OPTICAL FLOW AND EGO-MOTION

Interpreting apparent motion without overclaiming

Optical flow refers to apparent pixel motion between frames. In moving-camera footage, such motion can arise from camera translation, rotation and independently moving subjects; it is not synonymous with measured body velocity.



Approach / 00:00.8



Advance / 00:02.0



Reorientation / 00:03.2

OBSERVABLE PARALLAX AND MOTION CUES

As the wearer advances, nearby parked cars and pavement edges exhibit greater apparent displacement than distant buildings. Simultaneously, the person ahead changes screen position and scale. These cues support a reading of forward ego-motion and changing viewing direction.

What the clip demonstrates

Qualitative consistency among visual path, framing changes, near-field displacement and subject trajectory.

What this study does not assert

No numerical optical-flow field, stabilization metric, inertial sensor record or ground-truth camera pose was computed or supplied.

EVALUATION RELEVANCE

For model evaluation, optical-flow estimates could later be applied to quantify motion discontinuities. In this portfolio, the term describes the visible image-motion cues and a relevant analysis method - not a completed measurement.

CAPTURE PRESENTATION AND MEDIA FACTS

Screen graphics are not camera metadata

Recording-style graphics support recognizable bodycam presentation. Their visible styling must be separated from file metadata and claims about a real recording device.



Clip 01 / source frame: display-style timestamp, device text, a yellow triangular icon and peripheral image darkening.

On-screen visual elements

The clips include a top-right date/time and device-style overlay with a yellow triangular mark; one clip also shows a top-left device wordmark. Their consistent placement is a screen-presentation characteristic.

Verified file characteristics

Three supplied MP4 files, each approximately 6.04 seconds. Main video streams: H.264, 752 x 416 pixels, 24 frames/s. Audio streams are present, but audio is not evaluated in this visual case study.

SOURCE INTERPRETATION

The displayed dates differ between clips. These scenes are independent synthetic demonstrations, and the graphic dates, model identifiers and icons are not proof of camera make, recording date, location or police agency.

FINISHING CRITERIA

When such footage is prepared for an editorial production, an on-screen display check covers character stability, layout, edge clearance, legibility and whether the graphic remains fixed in screen coordinates during camera movement.

VISUAL QUALITY ASSESSMENT

A structured review of the rendered results

The assessment uses observable video evidence and a defined technical brief. It separates deliberate capture-style characteristics from unwanted generation artifacts.

01 / Camera-origin fidelity

Does the viewpoint behave like a body-mounted recording position rather than a detached drone, tripod or stabilized tracking camera?

02 / Motion coherence

Are forward translation, footstep-related oscillation, turns and settling behavior plausible for the movement represented?

03 / Scene geometry

Do road edges, sidewalks, house fronts, trees, vehicles and relative object scale remain spatially credible across adjacent frames?

04 / Subject continuity

Does the moving figure remain identifiable across the pursuit as it passes parked vehicles and changes direction?

05 / Image and overlay quality

Are peripheral treatment, motion blur, exposure response and screen overlays coherent with the intended bodycam visual appearance?

06 / Brief adherence and preference

Among generated candidates, which result better follows the specified camera origin, movement, environment and subject-action requirements?

REVIEW STANDARD

This rubric supports qualitative review; no accuracy percentages, motion ground truth or automated benchmark scores are claimed.

RESULTS AND PROFESSIONAL JUDGMENT

What the three clips substantiate

The strongest outcome is a set of visually distinct outdoor BWC-style shots, including a faster running sequence in which viewpoint, motion direction and scene geography are linked.

Clip 01 / contextual activity

The camera surveys a residential street with nearby uniformed figures. Figure proximity and street-scale perspective make this a useful study of coverage and orientation.

Clip 02 / forward residential movement

The view evolves along the sidewalk while preserving recognizable houses, trees, vehicles and road alignment. Its emphasis is environmental context under moderate motion.

Clip 03 / fast pursuit-style motion

The camera follows a running figure as the route turns toward a residential yard. Stronger viewpoint movement, near-field image shifts and subject travel distinguish this result.

Shared / recording-style presentation

Wide coverage, peripheral darkening, and date/time/device-like screen graphics reinforce the intended BWC aesthetic.

TARGETED FINISHING CHECKS

A delivery review would check fast-motion edge coherence, foreground-object stability and overlay character rendering, particularly around direction changes. These are routine conform checks for synthetic footage, not evidence that the central movement study failed.

The three clips are independent output studies. Continuity is assessed within each clip; identical officers, dates or locations across all three are not acceptance requirements.

INDUSTRY APPLICATIONS / PRODUCTION

Specific screen-production uses for dynamic BWC footage

The technical value lies in producing and evaluating camera-origin-specific material, not in replacing every live-action police scene.

POLICE PROCEDURALS / CRIME DRAMAS

Design fictional officer-perspective exterior approaches, rapid foot-pursuit inserts, neighborhood searches or intercut BWC viewpoints that match a scene's movement requirements.

ACTION PREVISUALIZATION AND STUNTVIS

Evaluate pursuit blocking, camera-wearer trajectory, sidewalk-to-yard transitions, subject visibility, safety-sensitive movement paths and shot duration before a physical setup.

DOCUMENTARY VISUALIZATION

Create clearly identified illustrative reconstruction shots when authentic body-worn footage is not available or should not be shown. Maintain a clean distinction between depiction and source evidence.

VFX / EDITORIAL PICKUP MATERIAL

Explore short synthetic POV inserts, alternate coverage or transitional movement shots that can be reviewed against a larger production's spatial and visual-continuity requirements.

VISUAL DEVELOPMENT

Provide comparative references for wide-angle coverage, motion amplitude, environmental scale and composited recording-display appearance to directors, editors or VFX supervisors.

These are credible production applications of the demonstrated capability, not claims of commissioned deployments or completed client cost savings.

PRODUCTION PLANNING AND AI EVALUATION

From motion previsualization to technical review

A body-worn chase insert poses practical decisions about performer movement, wearable rigs, location access, roadway control, nearby vehicles and continuity with adjoining footage.

Live-action inputs

For a physical shoot, the team may need performers, police-style wardrobe or approved insignia, prop control, body-mounted camera equipment, location agreements, safety planning and coordination.

Synthetic shot selection

For suitable brief inserts, generated material can explore path, pace, camera pose and composition before committing to a rig, a location or a full stunt setup. Real savings depend on scope and finishing demands.



Clip 03: a dynamic front-yard approach can serve as a previs reference for movement, camera visibility and route planning.

TRANSFER TO MULTIMODAL VIDEO ASSESSMENT

The same review practice supports generative-video evaluation: assess instruction adherence, temporal coherence, camera pose plausibility, scene geometry, artifacts and relative quality of candidate outputs. Findings can be recorded with frame times and concrete visual evidence for an AI production or model-quality team.

EVIDENCE STANDARD

This portfolio shows applied human visual judgment. It does not claim that these specific clips trained an AI system, improved model accuracy or replaced a measured filming budget.

PROFESSIONAL HANDOFF

What a production team can evaluate

My contribution is the ability to define a camera-specific visual objective, produce targeted examples and review the result against technical and editorial constraints.

Production brief

Shot intention, camera origin, environment, action path, subject framing, daylight condition and editorial disclosure status.

Synthetic coverage

Short BWC-style clips designed around wearer motion and scene behavior, delivered as individually reviewable takes.

Quality observations

Time-indexed notes on camera movement, spatial continuity, motion artifacts, frame legibility and on-screen graphics.

Editorial handoff

Approved selects, original files, still references, known limitations and intended use for previs, illustration or supplementary footage.



Source frame / pursuit study: movement direction, camera framing and residential geometry remain the key review objects.

DELIVERY QUALITY PRINCIPLE

Acceptance means the visual action is legible, the camera pose fits the intended wearable viewpoint, scene landmarks evolve plausibly and the result can be integrated into the specified editorial context after appropriate finishing.

PORTFOLIO NOTES AND SOURCES

Evidence, references and publication boundaries

A professional reconstruction should be specific about what is observable, what was designed and which applications are prospective.

SOURCE MATERIAL

Primary project source: three supplied MP4 files: Police Outdoor Operation1; Police Outdoor Residential Operation; PREBOFT CAM_5 (Resident Pursuit1). Source media inspected at original video resolution. Figures in this document are extracted from these files.

TERMINOLOGY AND INDUSTRY REFERENCES

Technical context: U.S. National Institute of Justice, *Primer on Body Worn Camera Technologies* (2016), and NIJ guidance on wide-angle field coverage and motion artifacts. These are external reference frameworks, not claims about a measured camera in the supplied footage.

Methodology: MIT *Foundations of Computer Vision*, Optical Flow Estimation chapter (definition of apparent interframe image motion).

Production context: Unreal Engine, *What Is Virtual Production?* (previsualization and stunt visualization).

nij.ojp.gov/library/publications/primer-body-worn-camera-technologies
visionbook.mit.edu/optical_flow.html | unrealengine.com/explainers/virtual-production/what-is-virtual-production

PUBLICATION AND METHODOLOGY BOUNDARY

All scenes are synthetic visual reconstructions for a technical portfolio. No actual incident, police recording, agency authorization or authentic device origin is asserted. Private prompt structures, proprietary production templates and exact generation workflows are not disclosed.

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